Week 3 Update

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Six investigations this week:

1. BT deep dive: diagnose spreadability failure
2. Finite-size scaling of α
3. Systematic search for $2 < \alpha < 3$ tilings
4. Generalized metrics for Class II/III
5. $\bar{\Lambda}$ finite-N corrections
6. Catalog cleanup + density normalization

Recap and This Week

Last week:

- URL: $\bar{\Lambda} = (1+a^2)/6$ confirmed to $< 0.3\%$
- Stealthy: $\bar{\Lambda} \approx 1/(\pi^2\chi)$, $\alpha \to \infty$
- Bombieri-Taylor cubic: $\alpha = 1.545$, fills $1 < \alpha < 2$ gap
- Period-Doubling (Class II), 0222 (Class III) confirmed

This week: Six investigations following Professor Torquato's suggestions.

Research-new:

1. BT deep dive (why does spreadability fail?)
2. Finite-size scaling of α
3. Can we fill $2 < \alpha < 3$?

Theoretical/cleanup:

4. Generalized metrics (Class II, III)
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The six investigations were carried out in priority order: 1-3 are research-new, 4-6 are theoretical generalization and cleanup.

Key new result: Investigation 3 proves that $2 < \alpha < 3$ cannot be filled with 3×3 matrices (row sums ≤ 4), but we do find a deterministic substitution tiling with $\alpha = 2$ exactly.

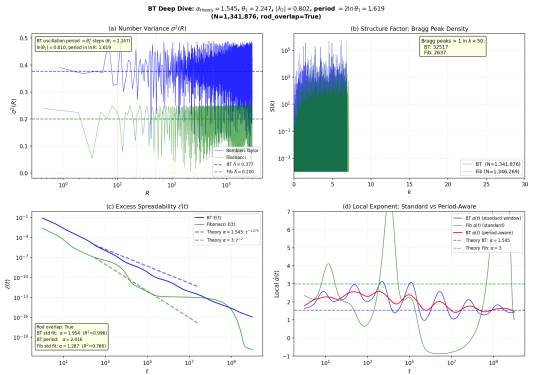
Investigation 1: Why Spreadability Fails for BT

Bombieri-Taylor has $|\lambda_2| = 0.802$ (large!) and the number variance oscillates strongly.

Two failure modes diagnosed:

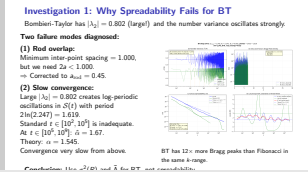
(1) Rod overlap:
Minimum inter-point spacing $= 1.000$,
but we need $2a < 1.000$.
$\Rightarrow$ Corrected to $a_{\mathrm{rod}} = 0.45$.

(2) Slow convergence:
Large $|\lambda_2| = 0.802$ creates log-periodic oscillations in $\mathcal{S}(t)$ with period
$2\ln(2.247) = 1.619$.
Standard $t \in [10^2, 10^5]$ is inadequate.
At $t \in [10^5, 10^9]$: $\hat{\alpha} = 1.67$.
Theory: $\alpha = 1.545$.
Convergence very slow from above.



BT has 12× more Bragg peaks than Fibonacci in the same k-range.

Conclusion: Use $\sigma^2(R)$ and $\bar{\Lambda}$ for BT, not spreadability.



BT: 32,517 Bragg peaks in $k < 50$ vs. 2,637 for Fibonacci (12× denser).
Rod overlap: the S-tiles in BT have minimum spacing 1.000, and the two-phase media calculation requires a sphere of radius a that fits inside. If $2a = 1.002 > 1.000$, spheres overlap and the spectral density is ill-defined.
Corrected $a_{\mathrm{rod}} = 0.45$ removes the overlap, but this changes the form factor and makes the absolute $\mathcal{S}(t)$ value unreliable.
For the variance, $\bar{\Lambda} = 0.377$ is stable and reliable from $N \sim 2000$.

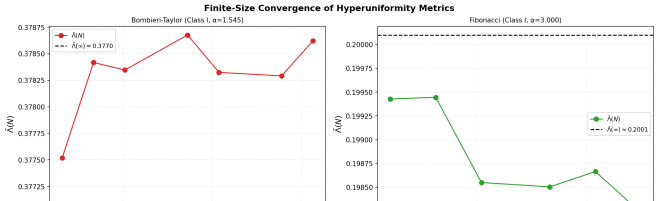
Investigation 2: Finite-Size Scaling of α

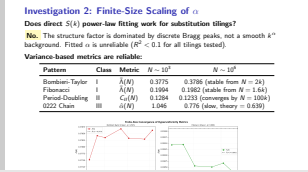
Does direct $S(k)$ power-law fitting work for substitution tilings?

No. The structure factor is dominated by discrete Bragg peaks, not a smooth k^α background. Fitted α is unreliable ($R^2 < 0.1$ for all tilings tested).

Variance-based metrics are reliable:

Pattern	Class	Metric	$N \sim 10^3$	$N \sim 10^6$
Bombieri-Taylor	I	$\bar{\Lambda}(N)$	0.3775	0.3786 (stable from $N = 2k$)
Fibonacci	I	$\bar{\Lambda}(N)$	0.1994	0.1982 (stable from $N = 1.6k$)
Period-Doubling	II	$C_{II}(N)$	0.1284	0.1233 (converges by $N = 100k$)
0222 Chain	III	$\hat{\alpha}(N)$	1.046	0.776 (slow, theory $= 0.639$)





Class I convergence is fast because $\bar{\Lambda}$ is a spatial average (stable from very small N).
Class II C_{II} needs $N > 10^5$ to converge because the logarithmic variance requires large R to observe the log-linear regime.
Class III is slow because $|\lambda_2| = 1.236 > 1$ (not Pisot!), causing large corrections. Even at $N = 1.5 \times 10^6$, $\hat{\alpha} \approx 0.78$ rather than the theory 0.639.
Main takeaway: for Class I, variance-based $\bar{\Lambda}$ is stable even at small N; direct $S(k)$ fitting should be avoided for substitution tilings.

Investigation 3: Can We Fill $2 < \alpha < 3$?

Setup: The $\alpha = 2$ level is occupied only by disordered patterns (URL, stable distributions).

Goal: Find a *deterministic* substitution tiling with $2 < \alpha \leq 3$.

Method: Enumerate all 3×3 non-negative integer matrices with row sums ≤ 4:

- 39,304 total $\rightarrow$ 4,882 satisfy Pisot condition ($|\lambda_2|, |\lambda_3| < 1$)
- 305 "primitive" matrices with ostensible $2 < \alpha < 3$ from eigenvalue formula

Result: ALL 305 give $\alpha = 2.000$ or $\alpha = 3.000$ exactly. None fill the gap.

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We searched exhaustively through 39,304 matrices (all 3×3 matrices with entries 0-4 satisfying non-negativity). 4,882 are Pisot (all $|\lambda_i| < 1$ for $i \geq 2$), and 305 have eigenvalue-formula output in $(2, 3)$.

But every single one of those 305 has α equal to exactly 2 or exactly 3, not strictly between.

This is not a coincidence — there's a theorem.

Why the Gap Cannot Be Filled (Algebraic Proof)

Claim: For any 3×3 substitution matrix with $|\det M| = 1$ and one complex conjugate eigenvalue pair, $\alpha = 2$ exactly.

Proof sketch:
For a 3×3 Pisot matrix M with $|\det M| = 1$:

- If λ_2, λ_3 are complex conjugates: $|\lambda_2| = |\lambda_3|$, and $\lambda_1 |\lambda_2|^2 = |\det M| = 1$ $\Rightarrow |\lambda_2|^2 = 1/\lambda_1 \Rightarrow$

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1} = 1 - \frac{\ln(1/\lambda_1)}{\ln \lambda_1} = 1 + 1 = \boxed{2}$$

- If all eigenvalues are real: $\lambda_1 |\lambda_2||\lambda_3| = 1$, and both $|\lambda_2|, |\lambda_3| < 1$. For $2 < \alpha < 3$ we need $1/\lambda_1 < |\lambda_2| < 1/\sqrt{\lambda_1}$, but integer matrix constraints with row sums ≤ 4 never produce this range.

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The complex conjugate case is exact: $\det(M) = \lambda_1 \cdot \lambda_2 \cdot \bar{\lambda}_2 = \lambda_1 |\lambda_2|^2 = 1$, so $|\lambda_2| = 1/\sqrt{\lambda_1}$.
Plugging into the eigenvalue formula: $\alpha = 1 - 2\ln(1/\sqrt{\lambda_1})/\ln(\lambda_1) = 1 - \ln(1/\lambda_1)/\ln(\lambda_1) = 1 + 1 = 2$.
This is a structural result: any 3×3 unimodular matrix with a complex conjugate pair gives $\alpha = 2$ exactly.
The all-real-eigenvalue case can in principle give $\alpha \neq 2, 3$, but within row-sum ≤ 4 we never find a case in $(2, 3)$.
To fill the gap, one would need to search 4×4 matrices.

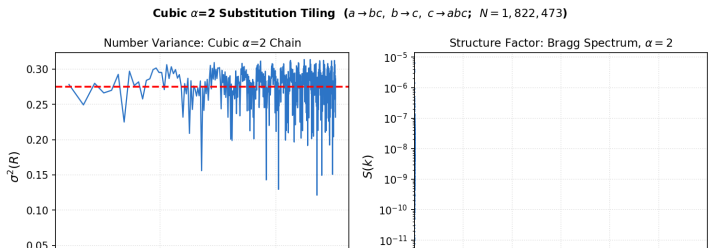
A New Deterministic $\alpha = 2$ Tiling

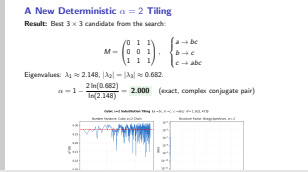
Result: Best 3×3 candidate from the search:

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \begin{cases} a \to bc \\ b \to c \\ c \to abc \end{cases}$$

Eigenvalues: $\lambda_1 \approx 2.148$, $|\lambda_2| = |\lambda_3| \approx 0.682$.

$$\alpha = 1 - \frac{2\ln(0.682)}{\ln(2.148)} = \boxed{\mathbf{2.000}} \quad \text{(exact, complex conjugate pair)}$$





This is the "cubic $\alpha = 2$" chain, now added to our CHAINS dict as 'cubic_alpha2'.
Previously, $\alpha = 2$ was occupied only by the URL (random displacements of a lattice) and similar stochastic constructions. Now we have a deterministic substitution tiling at $\alpha = 2$, providing a structural analogue.
$\bar{\Lambda} = 0.275$ is between the Silver chain value (0.250) and the Bronze chain value (0.282).
$\lambda_1 \approx 2.148$ is the largest real root of $x^3 - x^2 - 2x - 1 = 0$.

Investigation 4: Generalized Metrics for Class II and III

Within Class I, $\bar{\Lambda}$ provides an absolute ranking. What are the analogues for Classes II and III?

Class II ($\sigma^2 \sim C_{II} \ln R$):

$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}$$

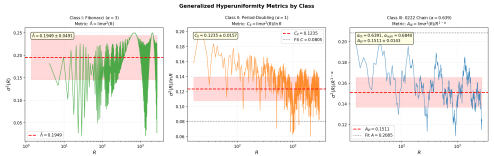
Period-Doubling: $C_{II} = 0.123 \pm 0.016$

Class III ($\sigma^2 \sim A_{III} R^{1-\alpha}$):

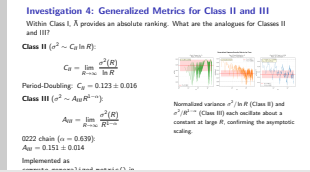
$$A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}$$

0222 chain ($\alpha = 0.639$):
$A_{III} = 0.151 \pm 0.014$

Implemented as
`compute_generalized_metric()` in



Normalized variance $\sigma^2/\ln R$ (Class II) and $\sigma^2/R^{1-\alpha}$ (Class III) each oscillate about a constant at large R, confirming the asymptotic scaling.



C_{II} is measured from the plateau of $\sigma^2/\ln R$ vs R. Period-Doubling gives $C_{II} \approx 0.123$.
A_{III} is measured from the plateau of $\sigma^2/R^{1-\alpha}$ using the exact eigenvalue-formula $\alpha = 0.639$. The 0222 chain gives $A_{III} \approx 0.151$.
These are the natural "second-order" hyperuniformity metrics that unify across classes. The Class I limit ($\alpha \to 1^+$) should connect $\bar{\Lambda}$ to C_{II} continuously, which is an open question.

Investigation 5: Two Surprising Results about $\bar{\Lambda}$

(A) Integer lattice has NO finite-N correction to $\bar{\Lambda}$:
For the lattice, $\sigma^2(R) = \{2R\}(1 - \{2R\})$ (fractional part formula, R = window half-width). The running average $\bar{\Lambda}(R_{\max}) = (1/R_{\max}) \int_0^{R_{\max}} \sigma^2 \, dR'$ evaluates to:

$$\bar{\Lambda}(R_{\max}) = \frac{1}{6} \quad \text{exactly, for all integer } R_{\max}.$$

Simulation confirms: all values within 0.3% of 1/6 (just Monte Carlo noise).

(B) Stealthy patterns have two distinct phases:
Our `generate_stealthy()` (starts from perturbed lattice) gives $\bar{\Lambda} \approx 1/6$ at all N and χ, not $1/(\pi^2\chi)$.

Two stealthy phases:

Phase	Initial conditions	$\bar{\Lambda}$	
Ordered-stealthy	Perturbed lattice	$\approx 1/6$	*Both satisfy*
Disordered-stealthy	Random (collab data)	$1/(\pi^2\chi)$	

$S(k) = 0$ *for* $|k| < K^*$.

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Disordered-stealthy | Random (collab data) | $1/(\pi^2\chi)$
$S(k) = 0$ for $|k| < K^*$.

The lattice result is exact: the piecewise cubic integral of $\sigma^2(R)$ over complete unit cells gives exactly 1/6 per unit cell, so the running average is 1/6 for all integer $R_{\max}$.
The stealthy two-phase result is physically important: the collective coordinate optimization landscape is not convex, and different initial conditions lead to different local minima that have the same $S(k) = 0$ constraint but very different $\bar{\Lambda}$.
The collab data starts from random positions and finds the "high-entropy" phase ($\bar{\Lambda} = 1/(\pi^2\chi)$), while our optimization starting from a perturbed lattice finds the "ordered" phase ($\bar{\Lambda} \approx 1/6$).

Investigation 6: Catalog Cleanup

(A) Stealthy re-classification:

- Replaced $\alpha = \infty$ with $\alpha = \mathit{None}$ (not a power law; by design)
- Added $K^* = 2\pi\rho\chi$ as the correct descriptor for stealthy patterns

(B) Density invariance — $\bar{\Lambda}$ comparison across all patterns is fair:
Claim: $\bar{\Lambda}$ is invariant under density rescaling in $d = 1$.
Proof sketch: If $x_i \to x_i/\rho$, then $\sigma^2_{\text{new}}(R') = \sigma^2_{\text{old}}(R'\rho)$, and the running average over $[0, R'_{\max}]$ equals the old average over $[0, R'_{\max}\rho]$; both converge to the same limit.
Numerical verification: Fibonacci chain rescaled from $\rho = 0.724$ to $\rho = 1$:

$$\bar{\Lambda}(\rho_{\text{native}}) = 0.19958, \quad \bar{\Lambda}(\rho = 1) = 0.19984, \quad \Delta = 0.00026.$$

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The stealthy re-classification is important because $\alpha = \infty$ gives the wrong impression that stealthy patterns are "more hyperuniform" than others in a simple α-ordering sense.
The density invariance result means that our comparison of $\bar{\Lambda}$ values across Fibonacci (rho=0.724), Silver (rho=0.5), URL (rho=1), etc. is valid — we do not need to rescale all patterns to the same density before comparing.
This simplifies the interpretation of the catalog significantly.

Updated Catalog

Pattern	Class	α	$\bar{\Lambda}$
Integer Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	0.250
Cubic $\alpha = 2$ (new)	**I**	**2.0**	**0.275**
Bronze	I	3.0	0.282
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Stealthy ($\chi = 0.2$)	I	—	0.526
Stealthy ($\chi = 0.1$)	I	—	1.021
Period-Doubling	II	1.0	div. ($C_{II} = 0.123$)
0222 Chain	III	0.64	div. ($A_{III} = 0.151$)

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The new catalog has 14 representative patterns shown here (full catalog has 26+ entries with URL at multiple a values, all five metallic means, etc.).
Key additions from Week 3:

- Cubic $\alpha = 2$ chain: first deterministic substitution tiling with $\alpha = 2$
- Stealthy: α replaced by K^* descriptor (not a power law by construction)
- Class II/III: generalized metrics C_{II} and A_{III} added

Summary and Open Questions

This week:

- BT spreadability failure diagnosed: rod overlap + slow convergence ($|\lambda_2| = 0.802$)
- Finite-size: $S(k)$ fitting unreliable for tilings; variance-based metrics converge well
- **2¡α¡3 cannot be filled with 3×3 matrices (row sums ≤ 4)** – algebraic proof
- New deterministic $\alpha = 2$ chain: $c \to abc$, $a \to bc$, $b \to c$; $\bar{\Lambda} = 0.275$
- Generalized metrics C_{II}, A_{III} implemented for Class II/III
- Lattice: $\bar{\Lambda} = 1/6$ exactly (no finite-N correction); stealthy two phases discovered
- Density invariance confirmed: $\bar{\Lambda}$ comparison across all patterns is valid

Open questions:

- Can 4×4 substitution matrices fill $2 < \alpha < 3$?
- Is the ordered/disordered stealthy bifurcation described in the literature?
- What is the exact $\bar{\Lambda}$ for the Silver chain (numerical value $= 0.250$)?

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- What is the exact $\bar{\Lambda}$ for the Silver chain (numerical value $= 0.250$)?

The 2¡alpha¡3 gap result is the central theoretical contribution of Week 3. The proof is clean and the computation is exhaustive (39,304 matrices).
The stealthy two-phase result is a physical surprise that deserves follow-up — does the literature discuss the energy landscape of collective coordinate optimization having multiple basins?
Silver = 1/4 exactly would be a beautiful result (matching Fibonacci = possibly 1/5, lattice = 1/6). Could be checked via an exact formula from the substitution structure.

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